

Power functions

A power function is a coefficient times a variable raised to a fixed real power.

$$f(x) = kx^p$$

Is $f(x) = 3x$ squared a power function?

The variable's position

$$f(x) = x^3, \quad g(x) = 3^x$$

Which is a power function?

A negative power

$$f(x) = x^{-2}$$

What is the real domain of this power function?

A fractional power

$$f(x) = x^{1/2}$$

What is the real domain and range?

Even powers

$$f(x) = x^6$$

What happens to the output as x becomes very large in either direction?

Odd powers

$$f(x) = x^5$$

How do the left and right ends of the graph differ?

A negative coefficient

$$f(x) = -2x^4$$

What is the end behavior on both sides?

Arrow notation

$$x \rightarrow -\infty, \quad f(x) \rightarrow \infty$$

What does this statement say about a graph's left end?

Polynomial functions

A polynomial function is a finite sum of terms with nonnegative integer powers of the input.

Why is $x^2 + 1/x$ not a polynomial function?

Degree and leading coefficient

$$f(x) = 7 - 3x^5 + 2x^2$$

What are the degree and leading coefficient?

A constant polynomial

$$f(x) = -8$$

What is the degree of this nonzero constant polynomial?

Leading-term behavior

For large input magnitude, the leading term determines a polynomial's end behavior.

$$f(x) = 2x^4 - 9x^3 + 1$$

Which term determines the ends of its graph?

Even degree, positive coefficient

$$f(x) = 3x^6 - x^2 + 4$$

What is its end behavior in arrow notation?

Even degree, negative coefficient

$$f(x) = -x^4 + 8x + 2$$

What is its end behavior in arrow notation?

Odd degree, positive coefficient

$$f(x) = 5x^3 - 2x^2 + 1$$

What is its end behavior in arrow notation?

Odd degree, negative coefficient

$$f(x) = -2x^5 + 7x - 6$$

What is its end behavior in arrow notation?

Factored form and degree

$$f(x) = -(x + 2)^2(x - 1)^3$$

What are its degree, leading coefficient, and end behavior?

A zero versus an intercept

A real zero r of f satisfies $f(r) = 0$.

$$f(x) = x^3 - 4x$$

Which graph points correspond to its real zeros?

Turning-point limits

A degree n polynomial has at most $n - 1$ turning points.

What is the smallest possible degree for a polynomial with four turning points?

A polynomial model

A model for inventory t months after counting is:

$$I(t) = -t^3 + 12t + 40$$

Does its long-term end behavior make unrestricted real-world use reasonable?